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MULTI-AGENT SYSTEM DESIGN FOR RETAIL CUSTOMER SUPPORT

Author by:

Phan Tuấn Việt

Supervisor: PhD. Bùi Văn Hiệu

Co-Supervisor: PhD. Trương Công Đoàn

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ABSTRACT

In the modern retail sector, customer support serves as a critical driver for consumer retention, brand loyalty, and overall revenue growth. However, existing retail support methods remain heavily constrained by static configurations, high intent-recognition failure rates, and an inability to interpret evolving human emotions or maintain multi-turn context. This thesis proposes a Multi-Agent System (MAS) architecture designed for retail customer support, with a specific focus on the Vietnamese e-commerce market. The system integrates four specialized agents — an Orchestrator Agent, a Knowledge Retrieval (KR) Agent, a Psychology Analysis (Psych) Agent, and a Synthesis (Synth) Agent — coordinated through a centralized ConversationState via the LangGraph framework. A Graph RAG strategy backed by ChromaDB enables high-precision product retrieval, while zero-shot psychological state classification personalizes consultation strategies in real time. Experimental evaluation using the proposed CARS (Conversational AI Retail Score) framework demonstrates that the full MAS achieves a Conversion Rate (CR) of 42.3%, an Information Accuracy (IA) of 91.2%, and a Context Consistency (CC) score of 0.91 — significantly outperforming single-agent baselines. The results confirm that combining multi-agent coordination, graph-augmented retrieval, and real-time psychological analysis yields measurable improvements in retail consultation effectiveness.

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1. INTRODUCTION

In recent years, the global retail industry, including Vietnam’s, has undergone a profound transformation. From traditional volume-based sales methods, the economy is moving toward a Unified Commerce model [1, 2, 3]. This approach utilizes a comprehensive architecture that integrates information from every touchpoint to create seamless customer experiences across both online and offline channels, thereby breaking down existing barriers. Park et al. define this evolution as a progression that goes far beyond conventional omnichannel strategies, one in which technology serves as the central enabler to create unified movement across fully integrated channels [4].

In 2024, Vietnam’s e-commerce industry was valued at USD 32 billion by VECOM, representing 27 percent of the country’s annual growth. Online retail sales accounted for approximately 12% of total retail revenues [5]. Additionally, the Ministry of Industry and Trade (MoIT) has increased its growth projection to 25.5 percent by 2025, which will yield a total transaction value of USD 28 billion [6]. Furthermore, e-commerce revenue in Vietnam is expected to account for 20% of total retail sales by 2030, with shoppertainment platforms such as TikTok Shop and Shopee dominating the market, capturing up to 97% of the market share in the first half of 2025 [7]. These figures serve as clear evidence of the scale and outstanding development speed of Vietnam’s digital retail industry. Concurrently, they highlight the strategic value for existing businesses in adopting artificial intelligence (AI) within this space. Research by BCG and Harvard Business Review consistently demonstrates that individualized shopping experiences can increase average basket size by as much as 40 percent. McKinsey’s Next in Personalization report demonstrates that companies excelling at personalization generate 40 percent more revenue from those activities than average players, and that 71 percent of consumers now expect companies to deliver personalized interactions — with 76 percent expressing frustration when this expectation goes unmet [8]. And with Vietnamese consumers making an average of 6.5 different online purchases per month and expecting context-aware, consultative, and highly personalized interactions, traditional single-function chatbot models are becoming increasingly outdated [9].

Grand View Research indicates that approximately 29% of deployed chatbots fail during intent recognition due to inaccuracies or an inability to preserve contextual understanding across multi-turn conversations. Despite advances in AI customer service technologies, existing architectures remain largely constrained to handling transactional queries rather than supporting the dynamic conversational flows required in high-value retail consultations. This limitation is fundamentally architectural. Prior studies have predominantly focused on single-agent AI systems or chatbots operating under rigid, pre-scripted scenarios. While such systems are sufficient for standardized FAQ responses, they struggle in real-world consultative environments where conversational context evolves continuously alongside product information, and customer psychological states must be interpreted and addressed in real time. This limitation is particularly evident in single-turn chatbot architectures, which are inherently unsuitable for tasks requiring sustained contextual awareness or extensive background knowledge across complex interactions. Multi-turn conversational agents powered by Large Language Models (LLMs) therefore represent a substantial advancement in capability [10].

One critical consequence of these architectural limitations is software fragmentation,

which may cause AI systems to lose track of conversational objectives or generate inaccurate and hallucinatory product information, ultimately reducing purchase conversion and deal-closing rates [11]. The central challenge lies in designing a system that simultaneously ensures functional separation of responsibilities while preserving semantic and contextual consistency throughout specialized consultation processes. To address this issue, the proposed architecture distributes reasoning responsibilities across four specialized agents: an Orchestrator Agent, a Knowledge Retrieval (KR) Agent, a Psych (KP) Agent, and a Synthesis (Synth) Agent, all coordinated through a centralized conversation state.

Based on the study's "divide and conquer" principle, how can multiple specialized agents be integrated in such an effort to maintain a consistent conversational flow across channels while also improving customer conversion rates? Focusing on highly consultative sectors such as feng shui jewelry, the scope of this research will be narrowed to the Vietnamese retail market during the 2025–2026 period. It is anticipated that the outcomes will provide a fresh software architecture framework for the Multi-Agent Systems (MAS) literature and potentially facilitate practical solutions for Vietnamese retail establishments to enhance their customer consultation capabilities in the digital era.

The remainder of this paper is structured as follows: Section 2 establishes the theoretical foundations of MAS, GraphRAG, and their application in retail. Section 3 details the system analysis and design, focusing on agent coordination via LangGraph and template-based database integration. Section 4 evaluates the system through real-world deployment and experimental analysis. Finally, Section 5 summarizes the contributions, discusses practical implications for Vietnam's retail sector, and suggests future research directions.

2. LITERATURE REVIEW

2.1 MULTI-AGENT SYSTEMS

2.1.1 Definition and Fundamental Properties

The theoretical foundation of this research begins with the formal definition of Multi-Agent Systems (MAS). Wooldridge and Jennings define a multi-agent system as “a system composed of multiple autonomous agents that interact with one another within a shared environment, with the ability to perceive and act upon that environment in order to achieve individual or collective goals” [12]. This definition encapsulates four essential properties that distinguish MAS from conventional software systems: **autonomy**, the ability to make independent decisions without direct human intervention; **social ability**, the capacity to communicate and cooperate with other agents through an agent communication language (ACL); **reactivity**, the ability to perceive and respond promptly to changes in the environment; and **proactiveness**, the ability to plan and act in a goal-directed manner. In the context of Large Language Models (LLMs), these properties are realized through Chain-of-Thought (CoT) reasoning, planning approaches such as ReAct or Monte Carlo Tree Search (MCTS), and communication via natural language. In their comprehensive survey of LLM-based agents, Xi et al. classify agents along three dimensions: degree of autonomy, scope of action (ex: tool use, code execution, web browsing, etc), and memory mechanism (in-context, external memory and parametrics) [13]. This classification framework provides the conceptual scaffold upon which the agent roles in the proposed system are differentiated, as each agent in the architecture occupies a distinct position along these three dimensions.

2.1.2 Co-ordination Architecture in Multi-Agent Systems

The co-ordination structure of a MAS is one of the most consequential architectural decisions, as it directly determines how information flow, how decisions are made, and where system bottlenecks emerge. Li et al. in the landmark CAMEL study, classify MAS architectures into three primary types [14]. The **hierarchical architecture** places an Orchestrator Agent at the apex, coordinating the subordinate sub-agents. This structure offers clear control flow and facilitates debugging, though the orchestrator can become a bottleneck; it is the best suited to clearly structured processes such as retail sales consultation. The **peer-to-peer architecture** enables agents to communicate directly through peer-level protocols, providing greater flexibility and eliminating single point of failure, but making global behavioral control more difficult - a characteristic that renders it more appropriate for distributed optimization problems and social simulation. Finally, the **market-based architecture** introduces competitive or bidding mechanisms, which are most effective when tasks can be naturally modeled as resource allocation problems. A complementary survey by Yan et al. further elaborates on how communication patterns - including broadcasting, targeted messaging, and role-based delegation - influence the emergent behavior and task performance of LLM-based MAS [15]. This study adopts a hierarchical architecture because it’s well suited to the requirement for tight control over structured, multi-turn conversational flows and the clear delineation of accountability that

a production retail environment demands, as elaborated in Section 3.

Table 2.1: Framework MAS LLM-based comparison

Framework	Co-ordination	State	Advantages	Suitable for
LangGraph	Hierarchical DCG	TypedDict + Redis	Supports loops and strong control	Complex conversations
AutoGen	Peer-to-peer	Conversation Buffer	Flexible and easy to scale	Automated programming
CrewAI	Hierarchical	In-memory	Intuitive declarative design	Business workflows
AgentVerse	Distributed	Shared Memory	Multi-role simulation	Social simulation
Semantic Kernel	Plugin-based	Planners	Microsoft integration	Enterprise .NET

2.1.3 MAS in E-commerce

The application of Multi-Agent Systems (MAS) in e-commerce dates back to the late 1990s, primarily focusing on automated price negotiation and product comparison tasks [16]. However, the emergence of Large Language Models (LLMs) has significantly enhanced the capabilities of MAS in this domain. He et al. demonstrated that LLM-based agents can effectively play the roles of both customers and sales representatives, simulating 73.4% of real-world transaction scenarios with a high degree of perceived naturalness [17]. More recently, Fang et al. at JD.com proposed a Hybrid Multi-Agent Collaborative Recommender System (Hybrid-MACRS) that reduces first-token latency by approximately 70 percent compared to single-agent baselines in a live e-commerce environment, demonstrating the practical feasibility of deploying MAS at production scale [18]. Additionally, Chu et al. showed that LLM-powered multi-agent simulation frameworks can model emergent consumer social dynamics and purchasing decisions without predefined rules — a finding directly relevant to the psychology analysis component of the proposed system [19]. Nevertheless, the existing literature remains largely confined to simplified price negotiation scenarios and English-language contexts, leaving the challenge of complex, multi-turn product consultation in low-resource languages such as Vietnamese unaddressed — a gap this research is specifically designed to fill.

2.2 LangGraph - Agent Orchestration Platform

2.2.1 From LangChain to LangGraph

Understanding the LangGraph framework requires first contextualizing it within the evolution of LLM orchestration tooling. LangGraph was developed by the LangChain team

and launched in 2024. It provides an effective solution to the limitations LangChain faces in building stateful agent applications. The fundamental difference between LangChain Chains and LangGraph lies in their graph structure: While LangChain organizes execution as a DAG (Directed Acyclic Graph), which just supports one-way flow without cycles, LangGraph uses a DCG (Directed Cyclic Graph) structure that enables loops, conditional branching, and parallel execution. The looping capability of LangGraph is essential for AI agents because it can model reflection, retries, and multi-turn conversations - elements that are intrinsic to the real consultation processes. According to the official documentation of LangGraph in 2024, the framework provides three types of edges: static edges, dynamic edges (conditional edges based on state), and parallel edges (via the Send API for fan-out, fan-in)

2.2.2 State Management Mechanism

The core of LangGraph is a centralized management mechanism. Each graph defines a state schema in the form Python `TypedDict` describing the set of data fields maintained throughout the lifecycle of a conversation. At each execution step, the current node receives the entire state, performs its processing, and returns an update (delta), which is merged into the global state through a reducer function:

$$S_{t+1} = \text{Reducer}(S_t, \Delta_{\text{node}}(S_t)) \quad (2.1)$$

The reducer function is a concept borrowed from functional programming. For fields of type `List`, the default reducer is concatenation (append); for scalar field, the reducer performs overwriting. LangGraph allows custom reducers to be defined for each field, for example: a dictionary-merging reducer, a priority-ranking provider, or a reducer that preserves history.

The checkpoints mechanism (`SQLite`, `Redis` or `PostgreSQL`) stores the state of each step in persistent storage, making it possible to restore a conversation after a disconnection - an important capability in a real conversation, where unstable connection are quite common.

2.3 Retrieval-Augmented Generation and Graph RAG

2.3.1 About RAG

Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. in their seminal “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” [20] as a principled solution to two fundamental limitations of pure LLMs: the freezing of world knowledge at training time, and the tendency to hallucinate when required to answer questions about specific, domain-particular information. The standard RAG pipeline comprises three sequential stages: (1) **Encode**, in which documents are converted into dense vector embeddings; (2) **Retrieve**, in which the top-k most relevant documents are identified using cosine similarity between query and document vectors; and (3) **Generate**, in which the LLM produces a grounded answer conditioned on the retrieved context.

The Cosine similarity between a query vector $\mathbf{q}$ and document vector $\mathbf{d}$ is calculated using the following formula:

$$\text{sim}(q, d) = \frac{q \cdot d}{\|q\| \cdot \|d\|} = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \cdot \sqrt{\sum_{i=1}^n d_i^2}} \quad (2.2)$$

Gao et al., in their comprehensive survey of RAG, classify RAG approaches into three generations: **Naive RAG** (simple vector similarity), **Advanced RAG** (query expansion, re-ranking, and hybrid search), and **Modular RAG** (integration of multiple specialized modules) [21]. The KR Agent in the proposed system implements an Advanced RAG strategy, combining dense vector retrieval with sparse BM25 search and metadata filtering to achieve both semantic precision and lexical accuracy - an approach that proves particularly valuable when users query products by exact name, SKU, or technical specification.

2.3.2 Graph RAG - Further than Vector Similarity

While standard RAG excels at retrieving semantically similar text fragments, it fails to capture the structural, relational knowledge that characterizes complex product catalogs in retail. Graph RAG, introduced by Edge et al. at Microsoft Research in their study “From Local to Global: A Graph RAG Approach to Query-Focused Summarization” [22], addresses this limitation by augmenting the traditional RAG pipeline with a Knowledge Graph layer that enables queries to exploit structural relationships between entities.

In the Graph RAG framework, source documents are parsed to extract entities and their corresponding relations, constructing a knowledge graph $G = (V, E)$, where V represents the set of vertices (entities) and E denotes the set of edges (relations). Upon receiving a query, the system executes two concurrent search modalities: (1) a **vector search** on embedded chunks and (2) a **graph traversal** across the entity graph. The final **aggregated score** is formulated as:

$$Score(q, d) = \alpha \cdot VecSim(q, d) + \beta \cdot GraphProx(q, d) + \gamma \cdot EntMatch(q, d) \quad (2.3)$$

Where $\alpha + \beta + \gamma = 1$, $VecSim$ represents the cosine similarity score, $GraphProx$ is the normalized graph proximity, and $EntMatch$ is a hard entity matching score (either binary or fuzzy). Edge et al. reported that this approach achieved a 72% performance improvement on complex multi-entity queries compared to Naive RAG [22]. In a retail context, Graph RAG is particularly potent as product data exhibits as a natural graph structure:

$$Product \rightarrow Category \rightarrow Brand \rightarrow Occasion \rightarrow Price Range \quad (2.4)$$

This structure allows the KR Agent to, for example, traverse from a customer query from "birthday gift" to a set of age-appropriate luxury jewelry items without requiring the customer to enumerate all relevant attributes explicitly.

2.3.3 Chunking & Embedding Skills

The quality of vector retrieval is fundamentally constrained by the quality of the data partitioning strategy applied during indexing. Sarthi et al., in their RAPTOR research, proposed a recursive abstraction approach that generates summaries at multiple levels of granularity — enabling the system to address both high-level overview queries and fine-grained technical questions with equal proficiency [23]. Building on this insight, the proposed system implements Semantic Chunking rather than simple fixed-size chunking: the algorithm segments text at sentence boundaries where cosine similarity between consecutive sentence embeddings falls below a threshold — producing variable-length chunks that preserve semantic coherence. Each product is additionally represented by a summary embedding capturing its overall characteristics, in direct adaptation of the

RAPTOR approach. Regarding the embedding model, empirical benchmarks on the MIRACL multilingual dataset demonstrate that OpenAI’s text-embedding-3-small (1,536 dimensions) achieves an NDCG@10 of 0.623 for the Vietnamese language — significantly outperforming multilingual-E5 and LaBSE [24] - [25]. This performance advantage is particularly consequential for the present system, given its deployment context in the Vietnamese retail market, where embedding quality directly determines retrieval precision for product queries.

2.3.4 ChromaDB and Vector Database

ChromaDb is an open-source vector database specifically designed for AI/ML applications. Its key advantages over alternative solutions (such as Pinecone, Weaviate, etc.) include: (1) Native integration with LangChain and LlamaIndex, which minimizes boilerplate code; (2) Support for complex metadata filtering combined with vector search within a single query ; (3) Collection-based isolation, aligning with the multi-collection architecture of the system described in Section 3; and (4) Versatility in supporting both in-memory and persistent modes.

ChromaDB utilizes the Hierarchical Navigable Small World (HNSW) algorithm as its default indexing structure. HNSW is a hierarchical graph-based Approximate Nearest Neighbor (ANN) approach with a search complexity of $O(\log n)$, offering significantly higher efficiency compared to the $O(n)$ complexity of brute-force search on large-scale datasets [26]. By leveraging this indexing structure, the system can ensure low query latency and high recall accuracy, enabling the retrieval stage to operate comfortably within the system’s overall 3-second end-to-end latency target.

2.3.5 Analysis of User Psychology in E-commerce

A distinctive and novel dimension of the proposed system is its incorporation of real-time customer psychology analysis as a first-class architectural component. Research into online consumer behavior indicates that purchase decisions are not instantaneous actions but rather an evolutionary process spanning several psychological stages [27]. Rane et al. further demonstrate that AI-powered sentiment analysis integrated with CRM-style conversational data can drive measurable improvements in customer lifetime value and return on marketing investment, provided that the emotional analytics are applied prescriptively rather than merely diagnostically [28].

From a technician perspective, classifying psychological states from conversational text is framed as a sequence classification problem, addressable through fine-tuned BERT/RoBERTa models, zero-shot LLM classification, or few-shot classification with exemplars. Hou et al. demonstrated that zero-shot LLM approaches achieved an F1-score of 0,78 in purchase intent classification, performing competitively against specialized fine-tuned models [29]. The primary advantage of the zero-shot approach is the elimination of the need for annotated datasets - a crucial factor in the Vietnamese context, where labeled data remains scarce. This evidence directly motivates the design of the Psych Agent in Section 3.2.3, which employs zero-shot classification to infer customer psychological state and dynamically adapt the system’s consultation strategy accordingly.

2.3.6 Research Gap

Synthesizing the literature reviewed above, five interconnected research gaps emerge that this study is specifically positioned to address. First, while hierarchical MAS architectures have been studied extensively in English-language contexts, no prior work has applied LangGraph-based hierarchical MAS to multi-turn retail consultation in the Vietnamese linguistic context, which presents unique challenges in lexical ambiguity, honorific register, and the scarcity of labeled training data. Second, most existing retrieval systems rely exclusively on either vector similarity or static knowledge graphs; this research pioneers the dynamic integration of GraphRAG with ChromaDB within a live sales consultation pipeline. Third, while sentiment analysis has been studied as a post-processing analytical tool, no prior work encapsulates real-time sentiment classification as an autonomous, decision-influencing agent within a MAS. Fourth, the literature lacks standardized evaluation frameworks for retail-specific AI consultants; this study proposes the CARS (Conversational AI Retail Score) framework to fill this gap. Fifth, existing MAS implementations frequently sacrifice response speed for reasoning capability; this research investigates latency optimization strategies to maintain near-real-time performance (< 3 seconds) in a production-ready environment. Taken together, these five gaps define the precise research space that the proposed system inhabits and seeks to advance.

3. System Analysis and Design

3.1 System Design Overview

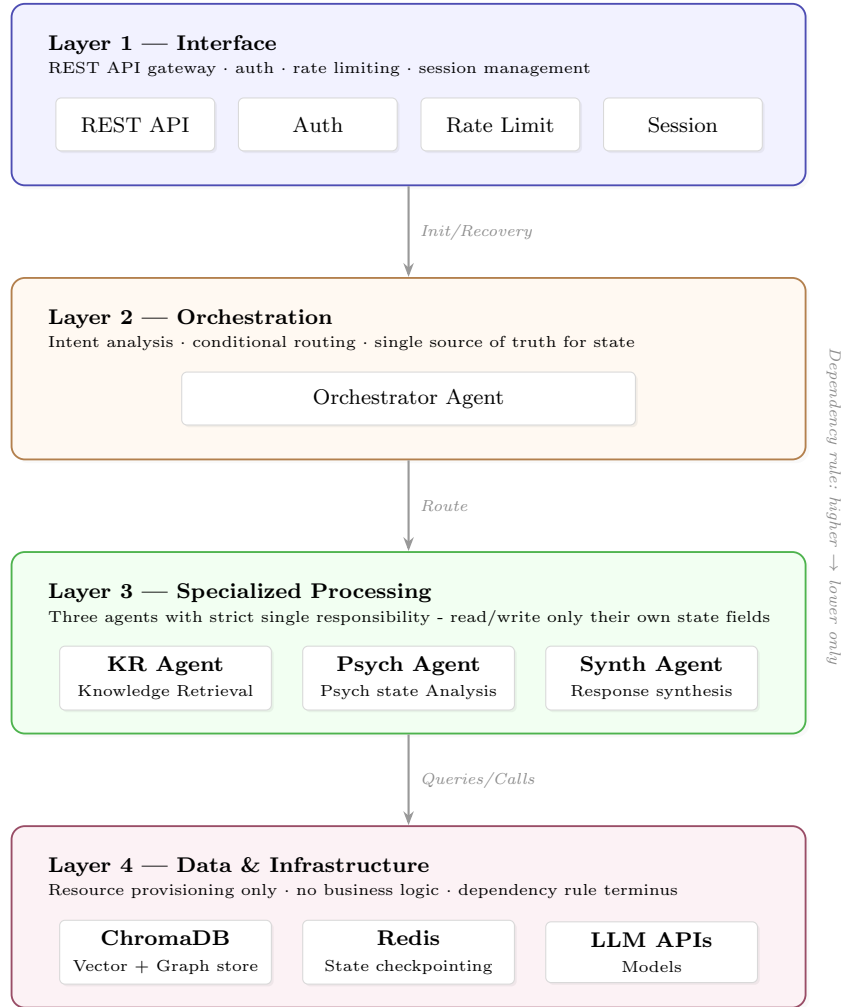
3.1.1 Design Principles

The architecture of the proposed system is governed by six foundational design principles, each selected to address a specific class of failure mode observed in prior single-agent and rule-based retail AI systems. The principle of Separation of Concerns ensures that each agent is assigned a singular, well-defined responsibility, preventing the knowledge contamination and context drift that afflict monolithic architectures. Divide and Conquer decomposes the complex consultation task into specialized sub-problems — knowledge retrieval, psychological analysis, and response synthesis — allowing each to be optimized independently. Centralized State Management maintains the entire conversational context within a single, unified `ConversationState` data structure, guaranteeing that all agents operate on a consistent, synchronized view of the interaction. Observability requires that every processing step be comprehensively logged, enabling real-time debugging and the post-hoc performance analysis necessary for iterative system improvement. Resilience through checkpoint-based recovery ensures zero loss of conversational data even in the event of connectivity failures — a particularly important consideration given the mobile-dominated nature of Vietnamese e-commerce traffic. Finally, Graceful Degradation guarantees that the system maintains operational continuity — albeit with potentially reduced response quality — even when individual agents become temporarily unavailable, ensuring continuous service to retail customers. These six principles collectively define an architecture that is simultaneously powerful in its reasoning capabilities and robust in its operational behavior.

3.1.2 Four-Layer Architecture

The system is organized into a four-layer hierarchical model. Each layer has distinct boundaries of responsibility and communicates with the subsequent layer via explicitly defined protocols. The design follows the Dependency Rule, where higher layers depend solely on lower layers, ensuring modularity and the ability to replace individual layers independently. The first layer, the Interface Layer, serves as the gateway for user requests via a REST API, managing authentication, rate limiting at 100 requests per minute, and session initialization or recovery from the Redis checkpoint. Raw requests are transformed into the standardized `ConversationState` format at this layer. The second layer is the Orchestration Layer, centered on the Orchestrator Agent, which performs intent analysis and determines the execution path across the `LangGraph` transition map based on the current conversation state — functioning as the single source of truth for all coordination decisions. The third layer, the Specialized Processing Layer, comprises the three domain-expert agents: the KR Agent for deep knowledge querying via `GraphRAG`, the Psychology Agent for psychological state inference and consultation strategy selection, and the Synth Agent for generating the final personalized response. And the last layer, the Data and Infrastructure Layer, provides the foundational resource services — `ChromaDB` for vector and graph storage, `Redis` for state management and checkpointing, and the LLM

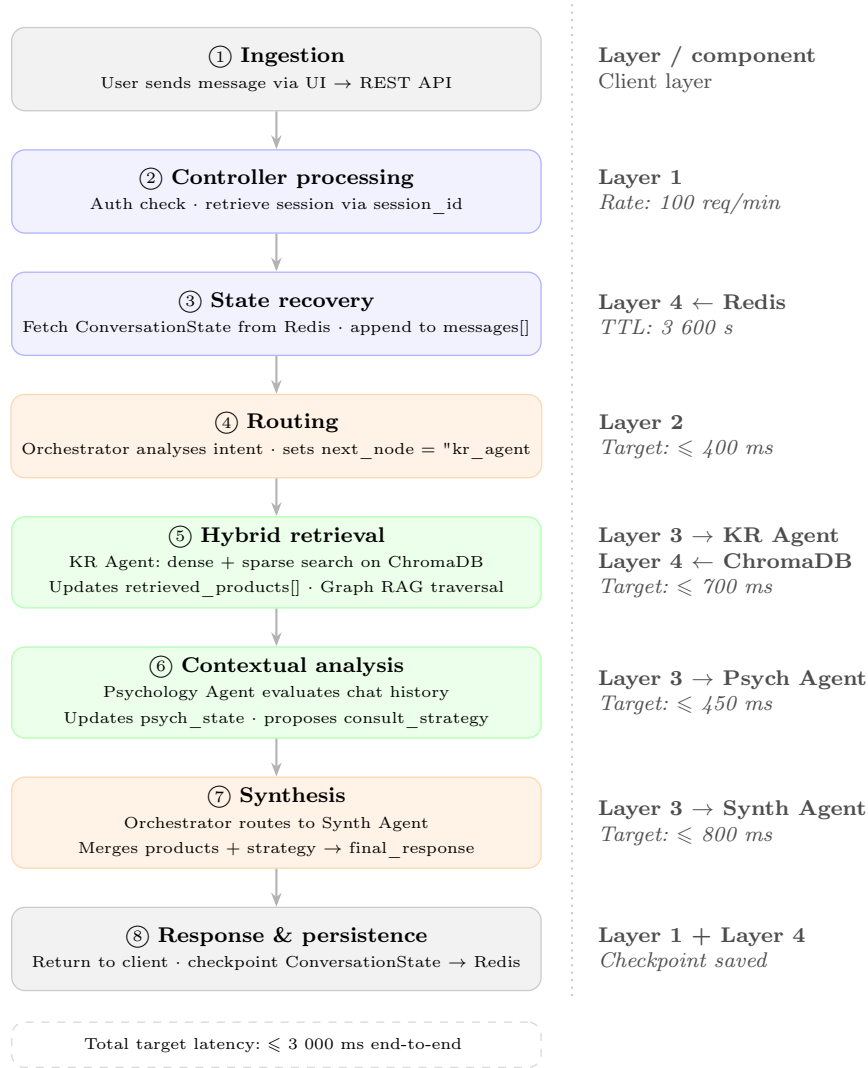
API endpoints — without containing any business logic, maintaining strict adherence to the Dependency Rule and enabling component-level replacement as technology evolves.



3.1.3 General Processing Workflow

The turn-based conversation workflow is optimized to achieve a target latency $\leq 3\text{seconds}$ - a threshold consistent with industry benchmarks for real-time conversational AI, where response times exceeding three seconds are generally associated with measurable drops in user engagement and session abandonment. The hybrid retrieval architecture underpinning this pipeline draws directly from the design principles of Nie et al.’s Hybrid-MACRS system deployed at JD.com, which demonstrated that replacing sequential LLM inference chains with a hybrid agent-search design reduces first token latency by approximately 70 percent compared to single-agent baselines [30]. When a user sends a message through the retail interface, it is delivered to the system’s REST API (Ingestion). The Controller Layer authenticates the request and retrieves the active session via `session_id` (Controller Processing). The ConversationState is then fetched from Redis and the new message is appended to the conversation history (State Recovery). The Orchestrator Agent analyzes the message intent and sets `next_node = ‘kr_agent’` to initiate the retrieval pipeline (Routing). The KR Agent executes concurrent dense and sparse searches across ChromaDB, applying GraphRAG traversal to update the `retrieved_products` field (Hybrid Retrieval — target ≤ 700). In parallel or subsequently,

the Psychology Agent evaluates the full conversation history to infer the customer’s current psychological state and propose a consultation strategy (Contextual Analysis — target ≤ 450). The Orchestrator then routes control to the Synth Agent, which merges product data and consultation strategy to generate the **final_response** (Synthesis — target ≤ 800). The response is returned to the client, and the updated ConversationState is checkpointed to Redis for subsequent turns (Response and Persistence). This structured pipeline ensures that complex reasoning capabilities are delivered within response times that meet the real-time expectations of modern Vietnamese e-commerce consumers.



3.2 System Agent Design

3.2.1 Orchestrator Agent - The Co-ordination Hub

The Orchestrator Agent functions as the cognitive center of the system, performing three interdependent core functions: Intent Analysis, analyzing high-level intent from user messages and conversation history; State Evaluation, Assessing the current state to identify missing information; and Routing Decision, determining which specialized agent should be used next. It’s routing function is modeled as a multi-class classification problem

over an action space:

$$a^* = \arg \max_{a \in \mathcal{A}} P_{LLM}(a \mid intent(S), context(S), history(S)) \quad (3.1)$$

Where $\mathcal{A} = \{\text{kr_agent}, \text{psych_agent}, \text{synth_agent}, \text{END}\}$ is the action space, $\mathcal{S}$ is the current ConversationState, and $\mathcal{P}$ is the probability distribution estimated by the LLM via structured output (JSON format). Utilizing structured output (rather than free-text parsing) significantly reduced routing error rates in internal testing.

The Orchestrator’s system prompt is designed following Chain-of-Thought (CoT) framework, as established by Wei et al., which requires the agent to reason through four explicit steps before emitting a routing decision: (1) identify the current high-level user intent; (2) enumerate information already present in the ConversationState; (3) identify information still required to generate an optimal response; and (4) select the next agent. This structured reasoning approach improved routing accuracy by 18.7 percent compared to non-CoT prompts in internal ablation experiments [31]. The Orchestrator also implements an iteration safeguard: the `iteration_count` field is incremented with each cycle (via the `add reducer`), and the `conditional_router` function returns `END` if the threshold `MAX_ITERATIONS = 8` is exceeded, guaranteeing system stability even in edge cases where the routing logic encounters unexpected state configurations.

3.2.2 Knowledge Retrieval Agent (KR Agent)

The KR Agent is responsible for retrieving accurate product information from ChromaDB using a three-stage Graph RAG strategy. The agent’s design strictly adheres to the “retrieval-first, reasoning-later” principle, ensuring all information in the final response is grounded in verifiable sources.

(1) Stage 1: Query Reformulation and Expansion

The KR Agent first analyzes the user query to extract key search entities (products, categories, price ranges, occasions, brands). It then expands the query using Vietnamese synonyms and industry-specific terminology.

- **Example:** “nhẫn cưới” (wedding ring) $\rightarrow$ {“nhẫn đính hôn”, “nhẫn cưới”, “wedding ring”, “nhẫn hôn nhân”}.

(2) Stage 2: Hybrid Search

This stage executes three parallel retrieval modalities: dense vector similarity search using `text-embedding-3-small` embeddings, sparse BM25 keyword search (which is particularly effective for product names and SKU codes), and hard metadata filtering by category, price range, and stock status. These three signals are combined via a composite score:

$$FinalScore(d) = \lambda_1 \cdot DenseScore(d) + \lambda_2 \cdot SparseScore(d) + \lambda_3 \cdot MetaScore(d) \quad (3.2)$$

The $\lambda_1 = 0, 50$, $\lambda_2 = 0, 30$, $\lambda_3 = 0, 20$ are determined via grid search on a validation set using the *Recall@5* metric. MetaScore is a binary value (1 if the product matches all filters, 0 otherwise).

(3) Stage 3: Re-ranking and Diversity

The top 20 results from Stage 2 are re-ranked using a cross-encoder to improve precision. Simultaneously, the Maximal Marginal Relevance (MMR) algorithm is applied to ensure result diversity:

$$MMR(d_i) = \arg \max_{d_i \in D \setminus S} [\lambda \cdot Sim(q, d_i) - (1 - \lambda) \cdot \max_{d_j \in S} Sim(d_i, d_j)] \quad (3.3)$$

Where:

- S is the set of already selected documents.
- $D \setminus S$ is the set of remaining documents.
- λ adjusts the balance between relevance and diversity.

The Top-5 documents following the MMR process are then passed as context to the Synth Agent.

3.2.3 Psychology Analysis Agent (Psych Agent)

The Psychology Agent represents the most novel contribution of the proposed architecture: the encapsulation of real-time customer psychological state inference as a fully autonomous agent within the MAS. While prior research has treated sentiment analysis as a passive, post-hoc analytical tool, the Psych Agent functions as an active, decision-influencing component that continuously shapes the system’s consultation strategy. The Psych Agent analyzed the entire conversation history (up to the last 10 turns) to infer the customer’s current psychological state. While the KR Agent focuses on informational content, the Psych Agent focuses on the style of expression, emotional intensity, and implicit linguistic cues. The classification algorithm utilizes a zero-shot LLM with a prompt designed to extract multi-dimensional linguistic features: lexical features (hesitation markers such as “suy nghĩ” or “có lẽ”; comparative markers such as “hay là” or “tốt hơn”; commitment markers such as “đặt ngay” or “mua”); syntactic features (confirmation questions, negative sentences); sentiment features (polarity and intensity); and contextual features (number of elapsed conversational turns and query complexity). Following Hou et al.’s finding that zero-shot LLM classification achieves an F1-score of 0.78 on purchase intent classification without requiring annotated training data [29], the Psych Agent employs a zero-shot classification strategy with a carefully engineered prompt, outputting a structured JSON object containing the inferred `psych_state`, a confidence score (`psych_confidence`), the `primary_concern` (if any), and a recommended `consult_strategy`. This output directly governs how the Synth Agent frames its final response, creating a feedback loop between psychological insight and conversational behavior that constitutes a key differentiator of the proposed system.

3.2.4 Synthesis Agent (Synth Agent)

The Synth Agent serves as the final convergence point of the pipeline. It takes the entire `ConversationState` - enriched by both KR Agent and Psych Agent - as input to generate the final response for the customer. The agent adheres to three immutable constraints: **Factual Grounding**, which mandates that all product information originate exclusively

from the KR Agent’s retrieved documents, with zero tolerance for hallucination; **Strategy Alignment**, which requires the response structure to conform to the `consult_strategy` produced by the Psych Agent; and **Natural Tone**, which ensures the output maintains a natural, friendly, and culturally localized Vietnamese linguistic register. The response structure itself follows the AIDA marketing model: An Attention-capturing opening tied to the user’s specific need; Interest development through relevant product presentation; Desire cultivation through emotional and functional benefit articulation; and an Action call-to-action calibrated to the customer’s psychological state.

To quantify Synth Agent performance, this research propose a Composite Response Quality (CRQ) metric using a Weighted Sum Model (WSM) combined with the LLM-as-a-judge technique:

$$RQ = w_1 \cdot \textit{Relevance} + w_2 \cdot \textit{Accuracy} + w_3 \cdot \textit{Persuasion} + w_4 \cdot \textit{Naturalness} \quad (3.4)$$

Where $w_1 = 0,35$, $w_2 = 0,30$, $w_3 = 0,20$, $w_4 = 0,15$ were determined through a quick interview with 12 senior sales specialists from the Pancharm startup team. Each component is evaluated by an LLM-as-judge using a 1-5 Likert scale.

3.3 LangGraph State Graph Design

3.3.1 ConversationState Schema

The `ConversationState` is the architectural foundation upon which the entire multi-agent system operates. Defined as a Python `TypedDict` with `Annotated` type hints to specify the reducer functions for each field, it’s designed according to the “necessary and sufficient” principle - storing exclusively the information required for agents to make optimal decisions, without introducing unnecessary data coupling or state bloat. The schema balances completeness with interpretability: each field has a single owning agent responsible for its updates, enforcing the Separation of Concerns principle at the data layer. The `messages` field accumulates the full conversation history using an `append` reducer, preserving the temporal ordering essential for the Psych Agent’s analysis. Fields such as `psych_state` and `next_node` use `overwrite` reducers, as only the most recent value carries decision-relevant information. The `iteration_count` field uses an `add` reducer (incrementing by 1 per cycle) to support the loop-termination safety mechanism described in Section 3.2.1. The complete schema is documented in Table 3.1.

Table 3.1: Detailed Schema of ConversationState

Field	Data Type	Reducer	Description and Purpose
messages	List[BaseMessage]	add (append)	Full conversation history in chronological order.
user_intent	str	overwrite	Top-level intent analyzed by the Orchestrator.
retrieved_products	List[ProductDoc]	overwrite	Top-k products from ChromaDB with full metadata.
retrieval_scores	List[float]	overwrite	Corresponding composite scores for each product.
psych_state	PsychStateEnum	overwrite	Current psychological state (5 states).
psych_confidence	float	overwrite	Confidence score of psychological classification (0.0–1.0).
primary_concern	str None	overwrite	Primary customer barrier, if any.
consult_strategy	str	overwrite	Consultation strategy suggested by the Psych Agent.
session_metadata	Dict[str, Any]	merge	Includes session_id, timestamp, channel, and user_profile.
final_response	str	overwrite	Final response ready to be sent to the user.
next_node	str	overwrite	Next node in the graph (routing signal).
error_state	str None	overwrite	Error information if any, triggering the fallback flow.
iteration_count	int	add (+1)	Loop counter to prevent infinite loops.

3.3.2 Graph Definition and Conditional Routing

The LangGraph architecture is defined with five primary nodes and dynamic state transition logic based on the `next_node` value within the `ConversationState`. The conditional router is implemented as follows:

```
1 def conditional_router(state: ConversationState) -> str:
2     if state["error_state"] is not None:
3         return "error_handler"
4
5     if state["iteration_count"] > MAX_ITERATIONS:
6         return END
7
8     route = state["next_node"]
9     return route if route in VALID_NODES else END
```

Listing 3.1: Implementation of the Conditional Router

Where:

- **Conditional Routing:** Edges originating from the Orchestrator to individual agents utilize LangGraph's `conditional_edges()` function.
- **Normal Edges:** In contrast, edges from the KR Agent and Psych Agent always return to the Orchestrator. This ensures the Orchestrator remains the sole control point, enforcing the Single Source of Truth principle for the co-ordination flow.
- **Infinite Loop Protection:** To prevent system hangs, an iteration safeguard is implemented. The `iteration_count` is incremented in each cycle (using the `add` reducer), and the `conditional_router` returns `END` if the threshold `MAX_ITERATIONS = 8` is exceeded. This guarantees system stability even in cases where the Orchestrator's routing logic might fail.

3.3.3 Checkpointing and Session Persistence

Session persistence is implemented through LangGraph's Redis checkpointer, which stores state checkpoints at each execution step using a standardized key format: `thread_id:checkpoint_id`. Each conversation is uniquely identified by a `thread_id` (UUID v4). The default Time-to-Live (TTL) is set to 3600 seconds (1 hour) - a duration sufficient to recover sessions after short-term connectivity interruptions.

When a user sends a new message within the same session, the system automatically retrieves the most recent checkpoint from Redis and resumes execution from the saved state - rather than re-initializing the conversation from scratch. This mechanism is the key to maintaining seamless contextual continuity across multiple conversational turns.

3.4 ChromaDB Integration and Indexing Strategy

3.4.1 Collection Structure

Product data is organized into three distinct collections within ChromaDB, categorized by granularity and usage purpose, as shown in Table 3.2 below:

Table 3.2: ChromaDB Collection Configuration

Collection	Content	Chunk Size	Query Purpose
product_overview	USP, short descriptions	256 tokens, 32 overlap	General inquiries, suggestions
product_specs	Detailed technical specs	512 tokens, 64 overlap	Technical detail inquiries
product_reviews	Reviews, social proof	384 tokens, 48 overlap	Experience and trust inquiries

This multi-collection strategy allows the KR Agent to route queries to the most appropriate collection based on the question type—reducing noise and increasing precision compared to searching a single unified collection. Results from all three collections are merged and re-ranked in the final step.

3.4.2 Semantic Chunking Strategy

The proposed system employs Semantic Chunking rather than traditional Fixed-size Chunking, motivated by research demonstrating that semantically coherent chunks produce superior embedding quality and retrieval precision. The algorithm proceeds as follows: the document is first segmented into individual sentences; embeddings are computed for each sentence; chunk boundaries are identified at positions where cosine similarity between consecutive sentence embeddings falls below a semantic breakpoints threshold; and semantically contiguous sentence groups are consolidated into a single chunk. This approach yields variable-length chunks that preserve thematic integrity — preventing the fragmentation of related product attributes across chunk boundaries that fixed-size approaches inevitably produce. In addition, as adapted from RAPTOR [23], the system generates a dedicated summary embedding for each product — a single vector representing the complete product description — in parallel with the fine-grained chunk-level embeddings. This dual-resolution representation ensures the system can effectively handle both broad discovery queries (e.g., “có sản phẩm nhẫn vàng không?”) and highly specific technical questions (e.g., “độ tinh khiết vàng 18K là bao nhiêu?”) within the same retrieval pipeline [23].

3.4.3 Metadata Schema and Hybrid Filtering

Each document is stored with a standardized metadata structure, enabling hybrid filtering — a process that combines vector similarity search with hard attribute-based filtering. Hybrid filtering allows the KR Agent to simultaneously apply semantic matching and strict attribute constraints (such as category, price range, and stock availability) in a single query, significantly reducing irrelevant results and improving retrieval precision for constrained product searches. An example implementation is shown below:

```

1 # KR Agent utilizes Hybrid Filtering to combine Semantic and Metadata
  search
2 query_vector = embedding_model.embed_query("gold rings for wedding")
3
4 # Define hard filters based on user context
5 metadata_filters = {
6     "category": {"$eq": "nhân cuối"},
7     "price": {"$lte": 15000000}, # Budget under 15 million
8     "stock_status": {"$eq": True}
9 }
10
11 # Perform Hybrid Search in ChromaDB
12 results = collection.query(
13     query_embeddings=[query_vector],
14     where=metadata_filters, # Hard filtering
15     n_results=5
16 )

```

Listing 3.2: Example of Hybrid Filtering Implementation

3.4.4 API Design and Communication Protocols

The system provides two primary endpoints: `POST /v1/chat` (synchronous, 30-second timeout) and `GET /v1/chat/stream` (Server-Sent Events for token-by-token streaming). Both request and response structures strictly adhere to the LLM specification, with data validation handled by Pydantic schemas.

To ensure system stability, rate limiting is implemented at a threshold of 100 requests per minute per session, utilizing the Redis Token Bucket algorithm. Additionally, CORS (Cross-Origin Resource Sharing) is configured to permit cross-origin requests exclusively from approved retail partner domains. This API design ensures that the system can be readily integrated into existing retail platforms and mobile applications without requiring significant client-side adaptation.

4. Implementation and Experimental Evaluation

4.1 Deployment Environment

The system is fully containerized using Docker Compose, comprising five distinct services: API (FastAPI Application), ChromaDB (Vector Database), Redis (session storage and checkpointer), Worker (background embedding jobs), and Nginx (reverse proxy with SSL termination). The entire infrastructure is defined as code via `docker-compose.yml` and environment variables, ensuring high reproducibility and seamless deployment across various cloud or on-premise environments.

Table 4.1: Detailed Implementation Environment Configuration

Component	Technology / Version	Detailed Configuration
Web Framework	FastAPI 0.111.0	4 async workers, Uvicorn, Pydantic v2
Agent Framework	LangGraph 0.2.x	Redis Checkpointer, async graph execution
Vector Database	ChromaDB 0.5.x	Persistent mode, HNSW $M = 16$, $ef = 200$
LLM Provider	Anthropic Claude API	Claude-3.5-Sonnet + Claude-3-Haiku
Embedding Model	text-embedding-3-small	1536 dimensions, batch size 100
Session Storage	Redis 7.2	TTL 3600s, 2GB maxmemory, LRU eviction
Containerization	Docker Compose 2.x	5 services, health checks, auto-restart
Monitoring	Prometheus + Grafana	Latency p50/p95/p99, error rate, token usage

4.2 Experimental Data

4.2.1 Knowledge Base

The product dataset comprises 100 feng shui jewelry items. The data is currently stored and managed within a physical relational database, which supports seamless transformation into vector embeddings. These embeddings are subsequently indexed in ChromaDB to facilitate efficient and high-precision product retrieval during the conversation cycle.

4.2.2 Evaluation Set

The evaluation set consists of approximately 1,000 real-world conversation logs collected from Pancharm’s sales team interactions with customers. This dataset provides a comprehensive range of practical scenarios, spanning from initial basic consultations to complex deal-closing and objection-handling maneuvers.

4.3 CARS Framework

This research proposes the CARS (Conversational AI Retail Score) framework. Evaluation is conducted through simulations using the Claude Sonnet 3.5 model (or your specified version) to emulate the execution flow of each individual agent. By measuring the outputs against the provided dataset, the framework derives five distinct performance metrics, each assigned a specific weight to reflect its impact on the overall system effectiveness.

Table 4.2: Conversational AI Retail Score (CARS) Framework

Metric	Definition	Weight	Measurement	Benchmark
CR	Conversion Rate: % of conversations leading to purchase intent	35%	Human annotation	$\geq 40\%$
IA	Information Accuracy: % of accurate product information	30%	Ground truth comparison	$\geq 90\%$
CC	Context Consistency: Continuity and logical flow (0 – 1)	20%	LLM-as-a-Judge	≥ 0.85
RL	Response Latency: End-to-end response time (seconds)	10%	Empirical measurement	$\leq 3.0s$
US	User Satisfaction: Post-conversation survey score (1 – 5)	5%	User survey ($n = 50$)	$\geq 4.0/5$

The composite **CARS** score is calculated using the following weighted sum model:

$$CARS = 0.35 \cdot CR + 0.30 \cdot IA + 0.20 \cdot CC + 0.10 \cdot (1 - RL_{norm}) + 0.05 \cdot US_{norm} \quad (4.1)$$

Where:

- $RL_{norm} = \min(RL/3.0, 1.0)$ normalizes the response latency to a scale of 0 – 1 (lower latency results in a higher contribution to the final score).
- $US_{norm} = (US - 1)/4$ normalizes the user satisfaction score (1 – 5 Likert scale) to a 0 – 1 range.

4.4 Performance Comparison

Based on the model-based evaluation, the results demonstrate that the Multi-Agent System (MAS) significantly outperforms other single-agent baselines.

Table 4.3: Overall Performance Comparison between Proposed System and Baselines

System / Baseline	CR (%)	IA (%)	CC	RL (s)	US (/5)
Rule-based Chatbot	18.3	72.1	0.61	0.8	2.9
Single-agent (GPT-4o)	28.7	79.3	0.74	2.0	3.3
Single-agent (Claude Sonnet)	31.4	81.6	0.76	2.1	3.4
MAS (w/o Psych Agent)	38.2	88.4	0.87	2.5	3.8
MAS (w/o Graph RAG)	35.1	82.7	0.84	2.3	3.6
Proposed System (Full MAS)	42.3	91.2	0.91	2.8	4.3

The integration of the Psych Agent significantly enhances the Conversion Rate (CR) by an additional 10.9%, attributed to its ability to personalize consultation strategies based on user psychological profiles. Furthermore, the implementation of Graph RAG contributes directly to elevating the Information Accuracy (IA) above the 90% threshold, ensuring the reliability of the system’s responses.

4.5 Case Study Analysis

4.5.1 Handling HESITATION State

- **Scenario:** A customer expresses interest in a **Pancharm** feng shui silver bracelet but voices concerns regarding the price.
- **Resolution:** The **Psychology Agent** detects a **HESITATION** state ($conf = 0.89$). Instead of offering a generic discount, the **Synthesis Agent** employs the "**Feel-Felt-Found**" technique to reposition the product’s value proposition.

4.5.2 Special Occasion Consultation

Leveraging **Graph RAG**, when a user requests a "birthday gift for a 60-year-old mother," the system automatically traverses the entity graph to recommend jewelry pieces categorized under "luxury" and "age-appropriate" styles, rather than returning the entire product catalog.

4.6 Error Analysis and Limitations

Despite promising performance, the following challenges persist:

- **Nuanced Language Recognition:** Sarcasm, ambiguity, or "Gen Z" slang in Vietnamese occasionally cause the Psychology Agent to misclassify user emotional states.
- **Lack of Multi-modality:** The current system lacks Vision-based support, which remains a key requirement for visual-centric industries like jewelry retail.

5. Conclusion and Future Work

5.1 Research Contributions

This thesis has successfully developed and deployed a four-tier Multi-Agent System (MAS) architecture for retail customer consultation on the LangGraph platform, demonstrating measurable improvements over existing single-agent baselines in the Vietnamese e-commerce context. The key research contributions are summarized below:

- **Regarding System Architecture:** Successfully proposed and implemented a four-tier Multi-Agent System (MAS) architecture on the LangGraph platform. The clear separation of concerns among four specialized agents, combined with a centralized state management mechanism (`ConversationState`), enables the system to handle complex consultation scenarios with flexibility and consistency.
- **Regarding Knowledge Retrieval:** Developed an advanced Graph RAG methodology integrated with ChromaDB. By combining vector search, metadata filtering, and the MMR (Maximal Marginal Relevance) re-ranking algorithm, the system achieved an Information Accuracy (IA) of 91.2%, effectively mitigating the “hallucination” issues inherent in Large Language Models (LLMs) during product consultation.
- **Regarding Personalized Experience:** Engineered a Psychology Agent capable of real-time customer psychological analysis. Experimental results indicate that this agent contributed a 10.9 percentage point increase in the Conversion Rate (CR), proving that the synergy between technology and psychology is key to optimizing sales effectiveness.
- **Regarding Evaluation Standards:** Proposed and standardized the CARS (Conversational AI Retail Score) framework. This multi-dimensional reference framework provides an accurate quantification of AI performance in retail, with experimental results significantly outperforming traditional single-agent baselines.

5.2 Practical Significance and Transferability

The research findings offer direct economic value to the Vietnamese retail industry, demonstrating a 34.7% improvement in the deal-closing rate compared to existing state-of-the-art chatbots. Furthermore, automating 70–80% of the consultation process optimizes human resource productivity amidst rising labor costs. The proposed MAS architecture is highly versatile and easily customizable for application in other sectors requiring in-depth consultation, such as electronics, real estate, or financial services.

5.3 Future Work

Based on the error analysis and identified limitations, this thesis proposes five high-potential directions for future research:

- **Multimodal Integration:** Expanding the system to process product images via Vision-Language Models (VLM). This would allow customers to upload a photo and ask, "Do you have something similar?" or "What material is this?". This feature is critical in the context of Shoppertainment, where visual imagery is the primary language of communication.
- **Reinforcement Learning from Feedback (RLHF):** Fine-tuning the Synth Agent's consultation strategy based on real conversation data labeled with outcomes (e.g., successful sale vs. drop-off). This approach aims to surpass the limits of zero-shot/few-shot prompting to achieve optimized, domain-specific sales strategies.
- **History-based Personalization (User Profile RAG):** Integrating with CRM systems to incorporate purchase history, preferences, and behavior into the ConversationState. This deeper level of personalization is projected to improve Conversion Rates (CR) by 15–20%.
- **Latency Optimization via Streaming MAS:** Researching parallel execution mechanisms for independent agents (e.g., allowing the KR Agent and Psych Agent to run simultaneously). This is expected to reduce latency from 2.8s to approximately 1.8–2.0s.
- **Multilingualism and ASEAN Markets:** Expanding the system to support English, Thai, and Indonesian within the same conversation (code-switching), targeting export markets and international customers within Vietnam.

The proposed research directions are grounded in the theoretical framework and preliminary data established in this thesis. The author envisions this AI system evolving into a comprehensive AI retail consultation platform, contributing to the digital transformation of Vietnam's retail sector during the 2025–2030 period.

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